Test File 5: Understand Caps and Floors

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1. About Caps and Floors

Caps and floors are known as protected interest rate contracts. A protected interest rate contract can be defined as a contract that for a fee grants the right to benefit fully from a rate move in one direction, while being exposed to only limited risk if the rate goes the opposite way.

The asymmetric nature of the contract leads to two types of contracts:

- **Cap.** The most popular is the interest rate Cap, which is used to set a maximum level, or Cap, on a short-term rate index (e.g., LIBOR). The purchaser of the Cap is compensated if the index goes above a certain level, known as the strike level.
- **Floor.** The other type of contract is the interest rate Floor, which sets a minimum rate on some index. An asset manager may purchase a Floor to guarantee some minimum return on a floating-rate asset. For a predetermined fee, the guarantor of the Floor would pay the purchaser whenever the index is below a certain level on a given set of dates.

For example, if the Strike Rate on a Floor contract is 5%, and on the reset date the underlying security coupon rate is 4.5%, then the Floor contract accrues at .5% (Strike Rate - underlying coupon rate). If the coupon rate on the underlying security is 5.5% on the reset date for the same Floor contract, then the security accrues at 0% because the underlying security coupon rate is greater than the Strike Rate.

Contrarily, if the Strike Rate on a Cap contract is 5%, and on the reset date the underlying security coupon rate is 4.5%, then the Cap contract accrues at 0% because the underlying security coupon rate is greater than the Strike Rate. If the coupon rate on the underlying security is 5.5% on the reset date for the same Cap contract, then the security accrues at .5% because the underlying security coupon rate is greater than the Strike Rate.

2. Set Up Entities for Caps and Floors

ABC Accounting has one field defined at the entity level that is specific to caps and floors. You must specify the Net Cap Floor field value prior to any trading of cap/floor contracts. If you do not specify this value prior to trading, trades do not process and tools such as the Book Trade tool do not display the correct trading options.

Field Name	Tag	Description
Net Cap Floor	5830	Determines how the system processes caps/floors. Options include: • **Yes.** The system nets a buy and sell of a cap/floor on the same entity into a single position. • **No.** The system treats a buy and sell of a cap/floor on the same entity as two separate positions. If you set the Net Cap Floor field to No, ensure you set the Tech Short Eligible Indicator (tag 57) at the entity level to indicate whether the position can go technically short.
Tech Short Eligible Indicator	57	Indicates whether the entity can go technically short (book sell transactions prior to buys that will eventually be posted). Options include: • **Yes.** Default. The system allows you to oversell, sell without a buy, and process a buy to cover without the short. Setting this field to Yes allows you to use the rollback and replay feature in ABC's Accounting solution to adjust earnings. The system creates a negative long position (for an oversell or sell without a buy) or a negative short position (for a buy to cover without the short) and when the buy is received, the system replays that sell transaction. If you enter Yes, the system creates a settlement as soon as the trade is processed. There is no trade date to settle date gain/loss. This applies even if the settlement date is in the future. • **No.** Recommended. The system does not allow you to oversell, sell without a buy, or process a buy to cover without the short. The system rejects the trade and sends it to the Exceptions workspace to be reprocessed manually.

For general information about entity setup and entity-level options, see Create/Edit Entity Panel Options and Create Master Fund Panel Options. Detailed information about the Net Cap Floor options follows.

2.1. About Treating the Cap/Floor as Two Separate Positions

If you set the entity's Net Cap Floor field to No, an entity can have both a long and a short position in the same cap/floor security master record. ABC Accounting stores separate long and short positions. Each cap/floor position is a either a long intent or short intent position.

Longs and shorts are *not* netted into one net position, and:

- A *positive* position in a *long* position quantity signifies a long position.
- A *negative* position in a *long* position signifies a technically short long position (short in a long intent position).
- A *positive* position in a *short* position quantity signifies a short position.

- A *negative* position in a *short* position signifies a technically short position (long in a short intent position).

If you set the entity's Net Cap Floor field to No, the entity election, Tech Short Eligible Indicator field value (tag 57), determines whether the position can go technically short. If you set Tech Short Eligible to Yes, the position can go technically short. Otherwise, if you set this option to No, the position cannot go technically short.

2.2. About Netting the Cap/Floor to a Single Position

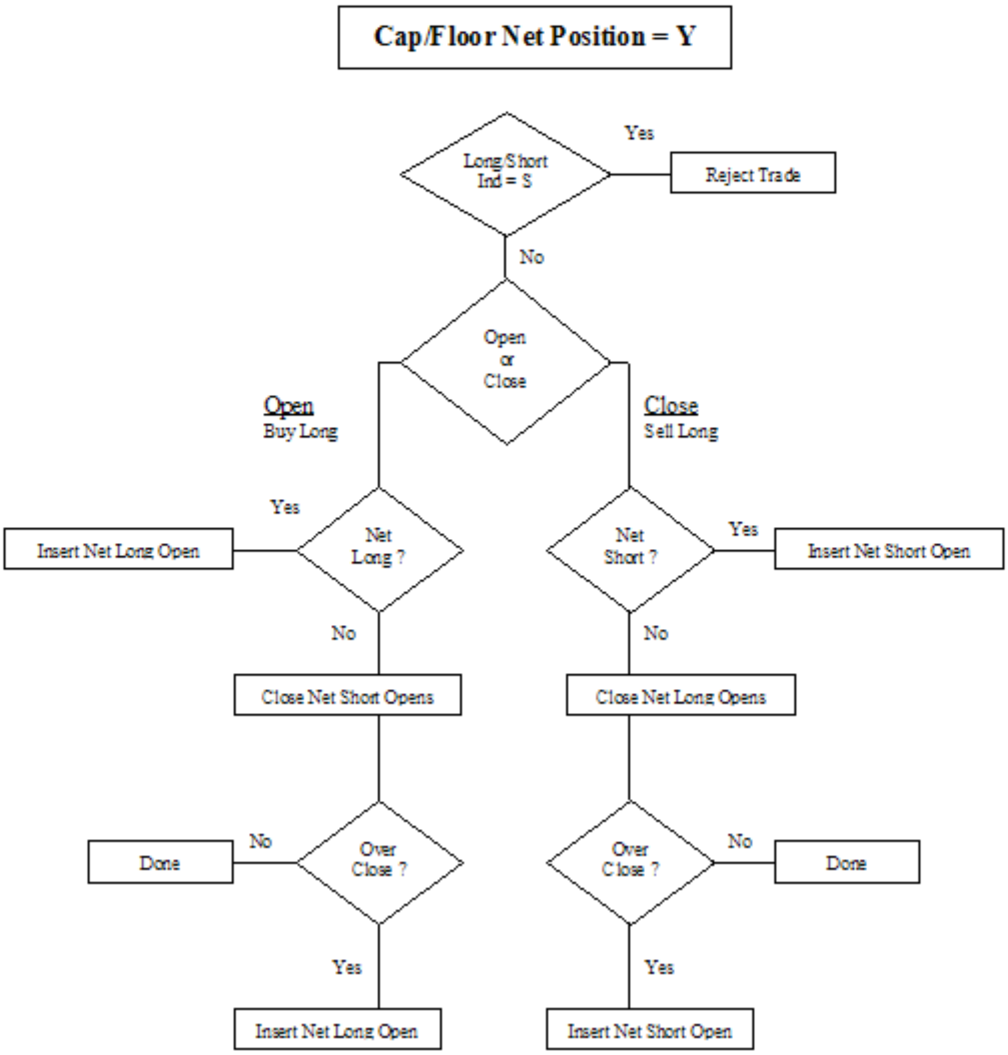
If you set the entity's Net Cap Floor field to Yes, ABC Accounting stores one net cap or floor position. An entity can be either net long or net short in the same cap/floor security master record, as illustrated in the following diagram.

Each Cap/Floor position is a net position. Longs and shorts are netted in one position, and:

- A *positive* position quantity signifies a net *long* position.
- A *negative* position signifies a net *short* position. The position is set up in ABC Accounting as a long position.

If you set the entity's Net Cap Floor field to Yes, a cap/floor position is always allowed to go technically short in the ABC Accounting long position. Setting the entity's Net Cap/Floor field to Yes overrides the entity's Tech Short Eligible Indicator field setting.

The following figure illustrates transaction processing for a cap/floor net position, where you set the entity's Net Cap Floor field to Yes.


Cap/Floor Net Position = Y
Long/Short Ind = S
Yes
Reject Trade
No
Open or Close
Open
Buy Long
Close
Sell Long
Yes
Net Long ?
Insert Net Long Open
No
Close Net Short Opens
Net Short ?
Yes
Insert Net Short Open
No
Close Net Long Opens
Over Close ?
No
Done
Yes
Insert Net Long Open
Over Close ?
No
Done
Yes
Insert Net Short Open

3. Set Up Securities for Caps and Floors

You can use Reference Data Center to establish a cap/floor Security Master File (SMF) record with the Cap - Floor data strategy. For more information, see Create and Delete Securities.

Otherwise, you can use Accounting Center's Issue Viewer, ABC's legacy security master tool. To set up a cap or floor security, you can use the Caps & Floors panel. For more information, see Manage Caps/Floors in Issue Viewer.

This section describes key options used to set up cap and floor securities. Several examples follow.

3.1. Caps and Floors Key Options

A cap/floor must adhere to the same requirements as all fixed income securities. For example, the First Coupon Date and Last Coupon Date values must be in sync with the payment frequency of the bond, and you must enter all required fields before submitting the record. Also, the values for the First Coupon Date, Last Coupon Date, and Maturity Date must not include Delay Days. For general information about reference setup, see Understand Fixed Income Reference Data.

> **Note:** ABC Accounting determines the payment frequency of the bond from the Payment Frequency, Interest Payment Timing, Business Day Calendar, Business Day Adjustment, Coupon Day of Month, and Day of Month Override fields.

When you set up a cap or floor security, review the following key options. The following options are available in Issue Viewer's Caps & Floors panel. Note options may vary according to your selections.

Field Name	Tag	Description
Processing Security Type	3931	Specifies the code value that the system uses to determine the type of cap/ floor you are adding. Options include: • **OPIRCA.** Interest Rate Option Caps. • **OPIRFL.** Interest Rate Option Floors.

Field Name	Tag	Description
Coupon Type Code	97	Indicates the type of coupon associated with the security. Options include: • **Floating Rate.** The security has fixed coupon dates and uses a variable rate that is based on an underlying index and index offset to calculate the coupon to use for earnings. You must enter the rate in the Variable Rate table. • **Variable Rate.** The security has fixed coupon dates and uses a variable rate based on the security identifier to calculate the coupon for use in earnings. You must enter the rate in the Variable Rate table. • **Inverse Floater.** The security has fixed coupon dates and uses a variable rate whose coupon rate is calculated inversely to the underlying index to which it is attached. When you select this value, STAR displays the following fields: Inverse Floater Rate, Inverse Floater Multiple, Underlying Issue Name, Underlying Asset ID, and Index Offset. You must enter the rate in the Variable Rate table. • **Step Coupon.** The security has fixed coupon dates and uses a variable rate based on the security identifier to calculate the coupon for use in earnings. You must enter the rate in the Variable Rate table. The system recognizes this option as a step bond (also called a step coupon bond, step up bond, or step down bond). • **Fixed Rate.** The security has fixed coupon dates and has a fixed coupon rate to calculate the coupon for use in earnings. • **Unscheduled Variable Rate.** The security has unscheduled payments and accrues interest based on a rate you enter in the Variable Rate table.
Strike Rate	11850	Determines the rate at which you can exercise against the cap/floor. This value stores the ceiling rate for cap contracts and stores the floor rate for floor contracts. If you set up the security with a Floating Rate or Inverse Floating Rate coupon type, ABC Accounting uses the value in the Strike Rate field along with the rate of the underlying security on the reset date to determine at what rate the contract accrues interest. If the coupon type on the cap/floor contract is not Floating or Inverse Floating, the Strike Rate field is for reference only.
Maturity Price	42	Specifies the price at which the security returns principal. The Maturity Price defaults to 0 because both cap and floor contracts are interest-only and there is no repayment of principal.
Trading Flat	3949	Indicates whether the security is trading with or without interest. As a general rule, cap and floor contracts trade without interest (trades flat). Therefore this field has a default value of Yes. In the event that a contract does trade with interest, you can change the value to No. Options include: • **Yes.** The system does not calculate accrued interest purchased or sold at the time of the acquisition or disposition. In the event of a disposition mid-coupon, the system creates a coupon at the end of the coupon period based on the interest earned during the period. • **No.** The system calculates accrued interest purchased or sold at the time of the acquisition or disposition.

Field Name	Tag	Description
OID Indicator	218	Indicates whether the security is OID (original issue discount) eligible. For caps/floors, because the Maturity Price (tag 42) of a cap/floor defaults to 0; the OID Indicator is set to No. Options include: • **Yes.** The security is OID eligible. For this flag to be set to Yes, the issue price must be less than the value in the Maturity Price field. The value in the Maturity Price field cannot be 0. • **No.** The security is not OID eligible.

3.2. About Coupon Setup for Caps and Floors

When you use the Caps & Floors panel to set up a cap or floor security, you must specify a Coupon Type (tag 97) field value. The following setup information is specific to the Caps & Floors panel.

ABC Accounting supports the following coupon types for cap/floor contracts.

- **Variable Rate.** If you select this value, you need to populate coupon rates in the Variable Rate table to accrue interest income. ABC Accounting uses the coupon rates directly from the Variable Rate table. The Strike Rate field is not used in the calculation of coupon rates.
- **Unscheduled Variable Rate.** If you select this value, you need to populate coupon rates in the Variable Rate table to accrue interest income. ABC Accounting uses the coupon rates directly from the Variable Rate table. The Strike Rate field is not used in the calculation of coupon rates.
- **Floating Rate.** If you select this value, ABC Accounting requires the following fields:
  - Underlying Issue Name (tag 1141)
  - Underlying Ticker (tag 1349)
  - Underlying Asset ID (tag 1348)
  - Underlying Security Asset ID (tag 1347)
  - First Rate Reset Date (tag 10911)
  - Reset Frequency Code (tag 1788)
  - Underlying Issue Name (tag 1141)
  For cap/floor security master records set up with a Floating Rate coupon type, ABC Accounting uses the coupon rates of the attached underlying security on the Rate Reset Date from the Variable Rate table. ABC Accounting then adds any applicable Index Offset (tag 215) to the coupon rate selected from the Variable Rate table. ABC Accounting then compares the sum of the underlying security Coupon Rate and Index Offset with the Strike Rate to determine what the coupon rate should be.
- **Inverse Floating Rate.** If you select this value, ABC Accounting requires the following fields:
  - Underlying Issue Name (tag 1141)
  - Underlying Ticker (tag 1349)
  - Underlying Asset ID (tag 1348)
  - Underlying Security Asset ID (tag 1347)
  - First Rate Reset Date (tag 10911)
  - Reset Frequency Code (tag 1788)

- Inverse Floater Rate (tag 1553)
- Inverse Floater Multiple (tag 4532)

3.3. Cap Coupon Rate Example

In this example, a Cap Security has:

- Strike Rate of 8%
- Underlying Index on the cap contract is LIBOR
- Index Offset is 250 basis points
- The security Resets and Pays coupons on January 1, April 1, July 1, and October 1.

Rates are as follows.

Rate Reset Date	Coupon Rate	Index Offset	Total Rate
1-Jan	6.0	2.5	8.5
1-Apr	5.0	2.5	7.5
1-Jul	5.5	2.5	8.0
1-Oct	4.5	2.5	7.0

On January 1, ABC Accounting compares the All in Rate (Total Rate), which is 8.5%, against the Strike Price of 8%. Because the security is a Cap contract and the All in Rate is greater than the Strike Price, ABC Accounting accrues the difference between the All in Rate and the Strike Rate. In this example, ABC Accounting accrues at a rate of .5% (8.5% - 8.0%).

In the following quarters, the rates are not greater than the Strike Rate, so ABC Accounting accrues at a Coupon Rate of 0.00%.

4. About Processing Trades for Caps and Floors

Before you can process a cap/floor contract, you must:

1. Set up an entity record with the Net Cap/Floor field set to a value of Yes or No.
2. Correctly set up a cap/floor Security Master File (SMF) record with all required fields populated and the coupon dates in sync with the payment frequency of the bond.
3. Set up an amortization rule to process cap or floor securities with an Amortization Method of Straight Line, Straight Line/Actual, or None.

After these steps are complete, an entity can process cap/floor securities transactions against established cap/floor SMFs. To process cap/floor transactions, you can use the Open and Close Caps/Floors panel that you can access using the Book Trade tool. For details, see Book Trades for Caps/Floors.

Depending on whether you set the entity's Net Cap Floor field to Yes or No, certain event types are available in the Book Trade tool's Open and Close Caps/Floors panel.

4.1. Open/Close Transaction Event Types Available when Entity's Net Cap Floor Option is No

If you set the entity's Net Cap Floor field to No, you can select the following values for the Event Type field (tag 55):

- BUY (Open Long)
- SELL (Close Long)
- WRITE (Open Short)
- BUYCVR (Close Short)

The value in the Event Type (tag 55) field that you select determines the trade's Long Short Indicator field (tag 15) value. An Open Long position is closed by a Close Long transaction. An Open Short position is closed by a Close Short position.

4.2. Open/Close Transaction Event Types Available when Entity's Net Cap Floor Option is Yes

If you set the entity's Net Cap Floor field to Yes, you can select the following values for the Event Type field (tag 55):

- BUY (Open Long)
- SELL (Close Long)

The Long/Short Indicator (tag 15) is set to Long when the Net Cap Floor field is set to Yes. An Open Long position is closed by a Close Long transaction. A Close Long position, that creates a technically short position, is closed by an Open Long position.

5. Understand Open Cap/Floor Transactions

When you open a cap/floor contract, the system performs the following processing:

- Calculates and stores the Local and Base Premium of the contracts. Some contracts do not have a premium paid or received (zero cost).
- Calculates any applicable traded interest.
- Creates an inventory position for the number of contracts. Buy to Open transactions are Long in inventory. Sell to Open (short sells) are Short in inventory.
- Uses the quantity of the open contract as the first period's notional amount to price and accrue interest on.
- Updates the general ledger with the cost of the contracts and the payable or receivable for the contracts.

The following information is specific to cap/floor processing:

- If a cap is sold short to open, the entity receives the premium, but it takes on the liability of paying out interest. If a floor is sold short, the entity receives the premium, but it takes on the liability of paying out interest.
- The premium paid or received for a cap contract is always paid in full at the start of the contract (up front). Premium for a cap or floor contract is never paid periodically over the life of the contract.
- The premium paid or received for a cap contract, or a floor contract is amortized as income or expense over the life of the contract.

5.1. Calculation for Open Cap/Floor Transaction

ABC Accounting executes the following calculation for an open cap/floor transaction:

Local Premium Value (the Cost/Proceeds) to Open:

*(Quantity * Quantity Scale) * (Price per Contract * Price Multiplier)*

Base Premium Value:

Local Premium Value / Spot Exchange Rate to Base

5.2. General Ledger Postings for Open Cap/Floor Transactions

When a contract is opened, the following general ledger postings are required.

5.2.1. For a Buy to Open

Local Currency = Settle Currency = Base Currency:

Debit/Credit	G/L Account Name	G/L Account Number
Debit	Cost of Investments	1010000100
Debit	Investment Interest Receivable	1007000500
Credit	Investment Payable	2002000100

Local Currency = Settle Currency ≠ Base Currency:

Debit/Credit	G/L Account Name	G/L Account Number
Debit	Cost of Investments (Local)	1010000100
Debit	Investment Interest Receivable (Local)	1007000500
Credit	Payable for Investments Purchased (Local)	2002000100
Debit	Cost of Investments (Base)	1010000100
Debit	Investment Interest Receivable (Base)	1007000500
Credit	Payable for Investments Purchased (Base)	2002000100

5.2.2. For a Sell to Open (Event Type = Write) or if Net Cap/Floor is set to Yes and Opening Transaction is a Sell

Local Currency = Settle Currency = Base Currency:

Debit/Credit	G/L Account Name	G/L Account Number
Debit	Investment Receivable	1002000100
Credit	Cost of Investments	1010000100
Credit	Investment Interest Payable	2004000400

Local Currency = Settle Currency≠ Base Currency:

Debit/Credit	G/L Account Name	G/L Account Number
Debit	Investment Receivable (Local)	1002000100
Credit	Cost of Investments (Local)	1010000100
Credit	Investment Interest Payable (Local)	2004000400
Debit	Investment Receivable (Base)	1002000100
Credit	Investment Interest Payable (Base)	2004000400
Credit	Cost of Investments (Base)	1010000100

5.3. Open Cap/Floor Transaction Example

Open Transaction Information:

- Buy to Open
- Trade Date 01/04/02
- Entity is GBP based
- Security is USD based
- Security Trades Flat

Option	Value
Broker	ABC Investment Systems
Trade Date	01/04/02
Settlement Date	01/04//02
Price Multiplier	.01
Quantity Scale	1.00
Quantity	1,000,000

Option	Value
Price	0.09575000
Spot Exchange Rate	1.455700000000 (GBP to USD) 01/04/02

Local Premium Value = (1,000,000 * 1.00) * (0.09575000 * .01) = 957.5 USD

Base Premium Value = 957.50 / 1.44557 = 657.76 GBP

General Ledger Entries executed by ABC Accounting on 01/04/02 follow.

Debit/Credit	G/L Account Name	Amount & Currency
Debit	Cost of Investments (Local)	957.50 USD
Credit	Payable for Investments Purchased (Local)	957.50 USD
Debit	Cost of Investments (Base)	657.76 GBP
Credit	Payable for Investments Purchased (Base)	657.76 GBP

When earnings processing is invoked, ABC Accounting posts to the following accounts.

Debit/Credit	G/L Account Name	G/L Account Number
Debit	Cap/Floor Amortization Expense	1002000100
Credit	Cost of Investments	1010000100
Debit	Receivable For Cap/Floor Interest	1007000590
Credit	Cap/Floor Interest Income	4001000190

Local Currency = Settle Currency ≠ Base Currency:

Debit/Credit	G/L Account Name	G/L Account Number
Debit	Cap/Floor Amortization Expense (Local)	1002000100
Credit	Cost of Investments (Local)	1010000100
Debit	Receivable For Cap/Floor Interest (Local)	1007000590
Credit	Cap/Floor Interest Income (Local)	4001000190
Debit	Cap/Floor Amortization Expense (Base)	1002000100
Credit	Cost of Investments (Base)	1010000100
Debit	Receivable For Cap/Floor Interest (Base)	1007000590
Credit	Cap/Floor Interest Income (Base)	4001000190

6. Perform a Notional Value Change for a Cap/Floor

In the Notional Value Change panel, you can process a notional value change for a cap/floor contract. You can increase or decrease the cap/floor contract's par value by a specified percentage.

To perform a notional value change for a cap/floor:

1. In Accounting Center, in the left navigation pane, click **Transactions** > **Trades** > **Book Trade/Rebook Trade** > **Book Trade.**
   You see the Book Trade workspace.
2. Complete the options in the Search Details pane to find the entity and cap/floor security that you want to update, and click **Search.**
   You see the search results based on the criteria you selected.
3. Select the row with the cap/floor security you want to update.
4. On the **Book Trade** tab, in the **Actions** group, click **Action Rules**, point to **Other,** and then click **Notional Value Change.**
   You see the Notional Value Change panel.
5. Under **Entity Information,** you must identify the entity.
6. Under **Issue Information,** you must enter the Trade Date and Settlement Date vales and identify the security.
7. Under **Accounting Information**, specify a value in the **Rate of Action** field.
   The Rate of Action (tag 1001) can be a positive or negative value. A negative value causes a decrease in the par value of the cap/floor contract. A positive value causes an increase in the par value. You enter the Rate of Action value in the decimal form of a percentage. E.g., you enter a 10% positive change in the number of shares of the position as .10, and you enter a negative 10% change in the position as -.10.
8. Click **Submit.**